

A Comparative Study of Genetic Algorithms, Simulated Annealing, and their Hybridization for Mixed-Integer Optimization Problems

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Metaheuristics | Optimization | Genetic algorithms | Simulated annealing
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Introduction

In real world exists several problems which can not be solved by exact methods due to restrictions or constraints, such as time, money, resources, complexity and etc. These problems can be found in almost all fields namely engineering, economics, networking, and medical applications therefor they need to be handled very carefully to get the best and most efficient solution. As well, it depends on the type of the problem to be solved considering that some of them need to be maximized or minimized (1).

For this reason, Metaheuristics Algorithms (MA) were introduced to solve different range of problems, particularly non-linear, non-convex, discontinuous, high-dimensional, combinatorial problems and whatever problem which can be categorized as non-deterministic polynomial-time (NP-Hard) (2).

Every year, numerous MA are developed and inspired by Evolutionary Algorithms (EA), Physical Processes (PP), Swarm Intelligence (SI) and Human Behavior (HB) (3), (4) However, the most popular and studied MA are Genetic Algorithms (GA) and Simulated Annealing (SA). GA is inspired by the adaptation of a living organism in its environment,

meanwhile the organisms more adapted would survive and reproduce, concurrently the less adapted would extinct, this idea was proposed by Charles Darwin in his book “*The origin of Species*” in 1859 (5). On the other hand, SA is inspired by...

Therefor this article presents the methodology of GA, SA and GA + SA, a description of the proposed problems, the experimental configuration, analysis of the results and finally the conclusions.

Benchmark Problems

To evaluate the performance of GA, SA, GA + SA, (6) suggests a set of problems to test out under different conditions and determine if the performance and efficiency is acceptable. Thus, a set of 3 problems were selected following the use of discrete, continuous and mixed variables, the problems are:

- Discrete (Combinatorial): Traveling Salesman Problem (TSP) (7)
- Continuous: Rastrigin function problem (8)
- Mixed: Knapsack problem

TSP. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

Rastrigin Function Problem. The function of Rastrigin is a non-convex, non-linear and multimodal function, being n-dimensional and denoted at the equation 1.

$$f(\mathbf{x}) = An + \sum_{i=1}^n [x_i^2 - A \cos(2\pi x_i)] \quad (1)$$

Where:

- $f(\mathbf{x})$ is the objective function to minimize

- $\mathbf{x} = (x_1, x_2, x_3, \dots, x_n) \in \mathbb{R}^n$ is a vector of variables
- A is a constant, usually $A = 10$, denoting the amplitude of oscillation
- n is the dimension of the vector $\mathbf{x}$
- $-5.12 \leq x_i \leq 5.12$ is the commonly used search space.
- The global minimum is at $\mathbf{x} = (0, 0, 0, \dots, 0)$ with a value of $f(\mathbf{x}) = 0$

The Rastrigin function is frequently used to evaluate the performance of optimization algorithms due to its complexity. The quadratic term x_i^2 creates a global minimum, while the term $-A \cos(2\pi x_i)$ introduces periodic oscillations with amplitude A . When combined with the quadratic term, these produce approximately 10^n local minima, determined by the oscillation frequency of the $\cos(2\pi x_i)$ term and the search space (9).

Consequently, the Rastrigin function poses a significant challenge in determining whether an algorithm can identify local minima without becoming trapped in them, thereby enabling it to locate the global minimum (10).

Knapsack Problem. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

Metaheuristic Algorithms

The main difference between Classic Algorithms (CA) and Metaheuristic Algorithms (MA) can be pointed out in quality solution, due to the CA can find an exact and optimal solution, meanwhile the MA can find an approximate and acceptable solution (2).

The results of the set of problems presented in the previous section are following these four criteria:

1. It does not guarantee the optimal solution, however it would find an acceptable solution and nearly optimal solution in a reasonable time.
2. An incorrect architecture of the MA can lead to premature convergence, which means an incorrect solution and wasted computational time.
3. The performance of any MA is excessive sensible to the configuration of it.
4. The computational time of any MA would be highly due to the number of iterations.

To solve the problems displayed in this article is strictly necessary to test out different MA, in consequence of the theorem of “No Free Lunch” which means that there is no better MA than another for a particular problem, thus various MA must be tested to find the best MA for the problem (3).

The four categories of MA are:

- Evolutionary Algorithms (EA): The theory of evolution by Darwin mentions that the most fit organisms can survive and reproduce, meanwhile the less fit organisms would extinct (11).
- Physical Processes (PP): The laws of physics and natural phenomena can be used as starting point to solve problems, where the gravity, temperature, radiation, etc. can be used as inspiration to solve problems (4).
- Swarm Intelligence (SI): It based by the nature and the behavior of the animals, where it tries to mimic the behaviour of them to achieve the supervivence of the species, by way of example, ants, wolfs, bees, birds, etc (3).
- Human Behavior (HB): The way of behave of humans across the society works as inspiration to develop new MA able to solve complex problems (4)

In this section, the most popular and studied MA are being presented, which are GA, SA and GA + SA.

Genetic Algorithms. Using the idea of principles of the evolution and the process of natural selection was proposed by John Holland with help of his students and colleagues at the University of Michigan (MIT) in the 1975 (12).

The generations of solutions must follow a process to guarantee a global solution, instead of a local solution, thus several operators must be applied to the population of solutions for each generation, which are:

- Selection: A probability of selection is applied to the fitness function for each individual, where the member with the best fitness value will be considered as the parents and the worst fitness value will be discarded.
- Crossover: It combines different segments of the parents’ genes to generate a new offspring, which will seek to achieve a better global solution.
- Mutation: A small random alteration is made to the offspring’s genes to increase population diversity, thereby avoiding a potential local solution and aiming to find a global solution through this genetic diversity.

Figure 1 shows a flowchart of GA. It illustrates how starting from an initial population, each individual is evaluated using the fitness function to determine its fitness value. Next, the termination condition is checked such as completing a certain number of generations or reaching an optimal score. If the condition is not met, selection, crossover, and mutation operators are applied to generate new populations until the termination condition is satisfied.

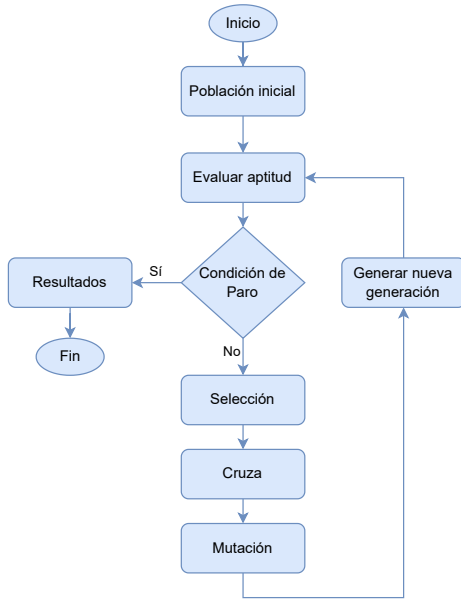


Fig. 1. Flowchart of a Genetic Algorithm (11)

Simulated Annealing. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

Genetic Algorithm + Simulated Annealing. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

Experimental Setup

Results

Just for kicks here’s a citation (1). And a reference to a supplement Sup. Note 1. And [Something about something](#). Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all!

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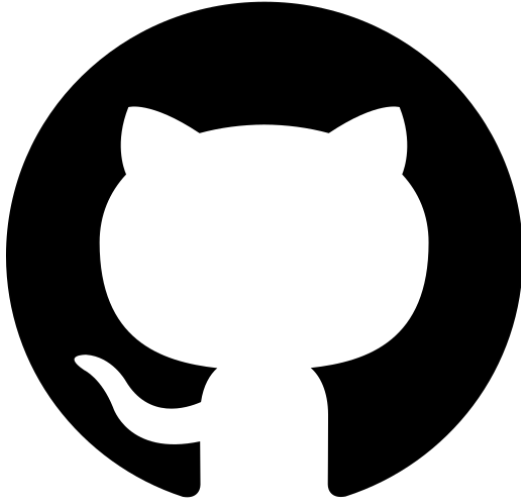


Fig. 2. Placeholder image with a long example caption to show justification setting.

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Figure 2 shows an example of how to insert a column-wide figure. To insert a figure wider than one column, please use the `\begin{figure*}...\end{figure*}` environment. Figures wider than one column should be sized to 11.4 cm or 17.8 cm wide. Use `\begin{SCfigure*}...\end{SCfigure*}` for a wide figure with side captions.

Table 1. Example results table, following the same layout used across my thesis chapters. Placeholder values.

Method	Metric	Value
Baseline	Accuracy	87.2%
Proposed	Accuracy	94.6%
Proposed	Latency (ms)	12.4

Table 1 shows an example of how to insert a results table using the same style as the thesis: a centered `tabular` with horizontal rules and blank spacer rows between entries.

Discussion

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Conclusions

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Future Work. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”?

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Supplementary Note 1: Something about something

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